Substance Misuse in Elite Athletes

Early Detection, Brief Intervention and Referral to Treatment

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KEYWORDS

• Substance misuse • Alcohol • Cannabis • Tobacco • Nicotine • Elite athletes • Early detection

KEY POINTS

- Encounters with alcohol, cannabis, and nicotine/tobacco misuse is common.
- Early identification can reduce the negative consequences of misuse and prevent the progression to a substance use disorder.
- The use of screening instruments, negative consequences questionnaires, and urine testing can aid early identification.
- Brief motivational interventions with personalized feedback, social norming, goal setting, contingency management, and significant other involvement can change attitudes, beliefs, and use patterns.
- The simultaneous use of alcohol and marijuana (SAM) among athletes is increasing and is associated with increased adverse effects.

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INTRODUCTION

Sports medicine physicians and athletic trainers regularly encounter athletes who misuse substances that put them at risk for adverse social, interpersonal, academic, psychological, health and performance effects.[1–6] The most often encountered substances in collegiate and professional athletes are alcohol (binge drinking), cannabis (marijuana), and nicotine (oral tobacco/nicotine vaping).[7–15] Male athletes typically misuse substances at higher rates than female athletes and male and female athletes from certain sports (eg, lacrosse, ice hockey, swimming, rugby) have the highest rates of use and misuse.[16–19] Differences in use and misuse rates of some substances in athletes versus non-athletes can begin in high school where, for example, binge drinking rates for athletes is higher than in non-athletes by 9th grade.[8]

Alcohol and other substance use typically begins as recreational or social use in high school and increases from 8th to 12th grades in both athlete and general student populations.[19,20] In 2022 a national survey demonstrated that past year use rates were significantly higher for 12th graders than 8th graders for alcohol (51.9% vs 15.2%), cannabis (30.7% vs 8.3%), and nicotine vaping (27.3% vs 12%). For some athletes, substance use becomes more regular and heavier, progressing to misuse by the end of high school. Misuse is defined as using alcohol or other drugs in a manner, situation, amount, or frequency that could cause harm to a user or to those around them.[21] Common adverse effects vary somewhat by substance class, but include physical, social, legal, emotional, performance and academic consequences[2,11,16] (**Table 1**). If substance misuse is identified early, prompt intervention can prevent progression to a substance disorder with or without comorbid mental health symptoms and disorders.

Substances may be used individually or concurrently with an increasing trend for using two together such that their psychogenic and behavioral effects overlap. The simultaneous use of alcohol and marijuana (SAM) and alcohol and tobacco/nicotine is common among college students including student athletes.[22–25] When two substances are used simultaneously, additive adverse effects such as heavier and more frequent use, driving while impaired, academic and athletic underperformance, and an increased probability of developing comorbid substance use and mental health disorders are likely.[23]

The early identification of athletes who are misusing any of these substances, alone or in combination, is critical to prevent the range of negative consequences and/or progression to a substance use disorder. Behavioral observations, the longitudinal use of screening instruments and/or negative consequences questionnaires, and random urine testing are important mitigation strategies.[1,2,4] Given the amount of time that sports medicine physicians and athletic trainers spend with athletes, they are in the best position to pick up early behavioral signs and symptoms of increasing substance use or misuse (see **Table 1**). The regular use of screening instruments throughout the year, and especially pre-season, should be considered. Short screeners include the three question Alcohol Use Disorders Identification Test-Consumption Questions (AUDIT-C Appendix 1), the four question Cutting Down, Annoyance by Criticism, Guilty Feeling, and Eye Opener Adapted to Include Drugs (CAGE-AID Appendix 2), and the eight or three question Cannabis Use Disorder Screening Tool Revised or Short Form (CUDIT-R, CUDIT-SF Appendix 3).[26,27] For those who screen positive and/or have a positive urine test, a more detailed assessment documenting the current frequency of use, route of administration, and negative consequences using either the Brief Young Adult Alcohol Consequences Questionnaire (B-YAACQ) or the Brief Marijuana Consequences Questionnaire (B-MACQ) should be conducted.[28–30]

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Table 1
Alcohol, cannabis, and tobacco/nicotine misuse: behavioral effects, intoxication, withdrawal, health and performance effects

Substance	Common Negative Behavioral Effects	Signs and Symptoms of Intoxication	Signs & Symptoms of Withdrawal	Negative Health Effects	Negative Performance Effects
Alcohol	Hangover, memory loss, passing out, embarrassing behaviors, lateness, low energy, interrupted/lost sleep, arguing or fighting missing class, arrested (fight, DUI), unwanted sexual encounter, smell of alcohol on the breath, irritability	Sedation, slurred speech, poor balance, unpredictable behavior, blackouts, memory problems, lack of attention, dizziness, vomiting, flushing, fights, arrests, sexual inappropriateness, mood swings, self-harm	Sweating, increased heart rate, hand tremors, insomnia, nausea/vomiting, agitation, anxiety, fatigue, irritability, clammy skin, dilated pupils, mood swings, confused thinking, headache, loss of appetite, depression, seizures	Reduced bone density, impaired fracture healing, pancreatitis, myopathy, wasting, respiratory distress syndrome, pneumonia, stroke, hypertension, heart disease, oral cancers, hepatitis, cirrhosis, liver, and colorectal cancers	Dehydration, insomnia, impaired motor skills, reduced glycogen resynthesis, increased injury rates, impaired temperature regulation, weight gain, hangovers, accidents, lateness, missing important events
Cannabis	Reduced motivation, decline in school performance, tiredness/ sluggishness, rudeness/ irritability, trouble sleeping, regret of behaviors, driving while high, derealization, depersonalization, anxiety, paranoia, social isolation	Euphoria, bloodshot eyes, lack of motivation, dry mouth, increased appetite, paranoia, anxiety, cough, slowed reaction time, lethargy, drowsiness, impaired judgment, social withdrawal	Irritability, anger, aggression, anxiety, nervousness, insomnia, disturbing dreams, decreased appetite, weight loss, depressed mood, restlessness, headaches, fever, chills, sweating, abdominal pain	Impaired short-term memory and learning, inattention, impaired motor coordination and driving skills, increased risk of injury, altered judgment, anxiety, paranoia, psychosis, addiction, altered brain development, quitting school, cognitive impairment, decreased life satisfaction and achievement, chronic bronchitis.	Increased heart rate and blood pressure, slowed reaction time, impaired motor coordination, impaired concentration, decreased motor agility, reduced mental quickness

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Table 1 (*continued*)

Substance	Common Negative Behavioral Effects	Signs and Symptoms of Intoxication	Signs & Symptoms of Withdrawal	Negative Health Effects	Negative Performance Effects
Tobacco/ Nicotine	External bulge in cheeks from moist snuff, spitting into a cup or on the ground, vaping pen in a locker or travel bag, using when ill or despite oral lesions, stained teeth and hands from nicotine, annoying other from smoking, vaping, spitting in public (eg, second-hand smoke or vapor)	Nausea, vomiting, headache, stomachache, loss of appetite, eye irritation, tremors, dizziness, anxiety, confusion, sweating, cough, increased heart rate, elevated blood pressure, excessive salivation	Irritability, anger, frustration, anxiety, cravings, restlessness, insomnia, depressed mood, difficulty concentrating, increased appetite, trouble sleeping	Heart disease, stroke, lung, gastrointestinal, and oral cancers, emphysema, chronic bronchitis, tooth loss, gum irritation, cataracts, rheumatoid arthritis, bone loss, chronic cough, erectile dysfunction, infertility, low birth weight babies, premature aging of the skin, delayed wound healing, decrease in taste and smell	Anxiety, insomnia, chronic respiratory infections, cough, shortness of breath, excess sputum production

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Once alcohol, cannabis, and/or nicotine/tobacco misuse is identified and an in-depth assessment has been conducted, various interventions such as psychoeducation, direct or web-based individualized feedback, goal setting, support system involvement, contingency management and/or brief motivational interventions can be considered.[31–34] If these interventions are not effective in reducing substance use and/or its adverse effects, then referral to more targeted treatment (eg, outpatient, intensive outpatient, residential, inpatient detoxification) may be necessary.

This article will focus on alcohol, cannabis, and tobacco misuse and explore early detection, brief interventions, and referral to treatment. Additionally, a brief description of the available levels of care for the treatment of severe or intractable substance misuse or a substance use disorder will be presented.

ALCOHOL

Background

Binge drinking, with a range of adverse effects, is commonly encountered in US collegiate and professional athletes with certain sports (eg, lacrosse, rugby, ice hockey, swimming, baseball, field hockey) exhibiting higher rates than the general population. Binge drinking is defined as 4 or more standard drinks per drinking occasion for females and 5 or more drinks for males. Super binge drinking is defined as 10 or more drinks per drinking occasion and is seen more often in US male collegiate athletes (13%) than in female collegiate athletes (1.8%). Binge and super binge drinking are typically defined by the presence of one or more binge drinking episodes over the past two weeks.[16] For male collegiate athletes, certain sports have high rates for binge drinking ranging from 50% for wrestling to 64% and 69% for ice hockey and lacrosse respectively. For females, the sports with the highest rates range from 43% for volleyball to 56% and 57% for ice hockey and lacrosse respectively.[16] The most endorsed negative effects for binge drinking among collegiate athletes are hangover (52%), loss of memory of events (28.4%), regret of behaviors (25.4%), interrupted/lost sleep (20.7%), got into an argument or a fight (16.8%) and missed a class (14.2%).[16] Even though binge drinking rates for US collegiate athletes from all sports have declined from 2009 to 2017 (58% to 44% for men; 51% to 39% for women), the rates are still high in certain sports and the adverse consequences are serious (ie, physical injury (6.8%), poor academic performance (7.1%), and/or poor athletic performance (7.4%)).[16]

Very limited data exist on the rates of alcohol misuse (including heavy use and binge drinking) among current professional athletes, especially in the US. The data that exists from Europe and Australasia suggests that in general, rates of alcohol misuse are highly variable depending on the sport.[18,35–39] For example, a study of professional soccer players from five European countries found rates of current alcohol misuse that ranged from 6.0% in Sweden to 15.7% and 17% for Spain and Finland, respectively.[35] A study of professional rugby players from Australasian countries ranged from 22% misuse rates in one study[36] to much higher rates of 62.8% in another.[37] Rates among current and retired professional ice hockey players from four European countries were 7.6% for current players and 28.7% for retired ones.[38] Somewhat higher current rates were seen in active professional cricket players from South Africa (26%), while retired professional cricketers had rates of 22%.[39] Both of these studies included male athletes only. However, a study of male and female professional soccer players in Australia found the rate of alcohol misuse in the women athletes (43.8%) to be lower than that of males (50.7%).[18]

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Early Detection

Athletic trainers and team physicians are in ideal positions to aid in the early detection of alcohol misuse. Proximity to, and time around athletes allows for frequent informal assessments and opportunities to learn about potential risky behaviors. Common behavioral signs of alcohol misuse that can be detected by athletic trainers, team physicians, and other support staff are detailed in **Table 1**.

Any athlete suspected of misusing alcohol or who has an adverse alcohol-related event (eg, blackout, fight, arrest, driving under the influence) should be assessed by the team physician or licensed mental health provider using motivational interviewing techniques and should receive the full 10-item Alcohol Use Disorders Inventory Test (AUDIT) adapted for use in the United States (USAAUDIT)[40,41] and the 24 item Brief Young Adult Alcohol Consequences Questionnaire.[42] USAAUDIT scores of 7 to 15 for women and 8 to 15 for men indicate hazardous drinking and should result in feedback and a brief intervention designed to reduce consumption and curb the adverse effects of alcohol misuse.

Another early detection strategy is to include alcohol urine testing as part of the panel for random screening of drugs of abuse and performance enhancers. Any level of alcohol in a morning urine test indicates heavy drinking the night before and should be assessed via motivational interviewing and administration of the USAAUDIT. Finally, because of the increasing likelihood of simultaneous use of alcohol and cannabis misuse, a cannabis use history (frequency, intensity, route of administration) and screening with the 8-item CUDIT-R or the 3-item CUDIT-SF (see Appendix 3) should occur. If SAM is present, there is an increased risk for comorbid mental health diagnoses and a screening for anxiety (Generalized Anxiety Disorder Screen-GAD-7), depression (Patient Health Questionnaire-PHQ-9), and sleep problems (Athlete Sleep Screening Questionnaire-ASSQ) should also be considered.[26]

Brief Interventions

Psychoeducation about the adverse effects of alcohol on sleep, energy, reaction time, emotional control, interpersonal relationships, and injury risk as well as its likelihood of reducing athletic and academic performance is a primary component of brief intervention strategies for alcohol misuse. Many athletes are unaware of the fixed metabolism of alcohol (only one drink an hour) which makes it unique among substances of misuse and increases the risk of heavy, rapid drinking. In addition, if alcohol is used simultaneously with cannabis or sedative/anxiety/depression medications (prescribed or over the counter) such as antihistamines, benzodiazepines, selective serotonin reuptake inhibitors (SSRIs), or sleeping pills, this potentiates the intoxicating effects of alcohol and makes adverse effects more likely.[1,2,4] Providing athletes with this information can help them make more educated decisions about their alcohol consumption.

For more serious alcohol misuse, as evidenced by a score of 16 to 24 on the USAAUDIT, a series of 1 to 4 brief motivational interviewing sessions should be implemented by a mental health provider working directly with the athlete/team or by an outpatient substance treatment program. These sessions should include personalized feedback, social norming, goal setting (academic, athletic, social), involvement of significant others (eg, peer, parent, coach), strategies for cutting down aimed at changing patterns of harmful use, and/or a trial of abstinence.

One example of such an intervention was created by Donohue and colleagues[32] for incoming first-year collegiate athletes. Following an assessment of the athletes' drinking behaviors, they created a one-session model (individual or group) with a follow-up assessment at 2 months that was led by a performance coach and included significant

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other involvement (typically a parent), goal setting and contingency contracting for goal achievement with an overall focus on alcohol as a potential barrier to sporting performance compared to a control group. Results showed that the brief intervention group, compared to controls, significantly reduced their alcohol consumption from baseline and experienced reductions in academic problems, injury concerns, dysfunctional thoughts, and stress.

Other interventions use on-line substance education modules (eg, *myPlaybook*) with personalized feedback and a focus on normative perceptions, behavioral expectancies, and harm prevention.[43] One such study using a revised version of *myPlaybook* and three iterative revision trials of 2000 to 3000 student athletes each was conducted by Rulison and colleagues.[34] While the first two trials did not result in reductions in alcohol and cannabis use at 30 days, the modified modules in the third trial had positive results. Results showed changes from baseline in normative views (ie, fewer athletes use substances regularly), positive alcohol expectancies (reduced), and negative marijuana expectancies (increased), but not on intentions to use behavioral strategies to reduce use.[34]

CANNABIS

Background

Self-reported rates of cannabis (marijuana) use and misuse among US collegiate and professional athletes are rising due to its expanded availability from medicalization/legalization and the decision by some professional sports (eg, Major League Baseball, National Football League) to reduce or discontinue random urine testing and eliminate sanctions in favor of evaluation and treatment.[10–12,16,44,45] Sports medicine providers should be aware of the prevalence rates, reasons for use, patterns of use, adverse effects, trends in potency, signs of misuse, and detection and intervention strategies for this substance.[2,4,6,16]

Studies of elite athletes in general show that approximately 1 in 4 use cannabis.[10,16,44] In US collegiate athletes, annual use rates are higher in: (1) men than women (26.3% vs 22.3%); (2) lacrosse, swimming, and wrestling for men (50%, 35%, 29% respectively), (3) lacrosse, ice hockey, field hockey, soccer and volleyball for women (34%, 29%, 29%, 27%, 27% respectively), (4) Division III than Division I (33% vs 18%) and in (5) states where it is legal than in those where it is not (39% vs 26%).[16] The most common reasons for use are socialization, pain control, well-being, stress reduction, and insomnia.[16,44]

Patterns of use range from episodic cannabidiol (CBD) alone and/or delta 9 tetrahydrocannabinol (THC with CBD simultaneously) to long-term regular use of higher potency THC with or without CBD and/or simultaneous use of alcohol and marijuana.[25,44] The most common routes of administration are inhaling (eg, smoking, vaping) or ingestion (edibles, tinctures)[16] and the most common short-term adverse behavioral effects are detailed in **Tables 1**.[46–48] When cannabis use becomes daily or near daily, the risk of additional adverse effects including the use of other illicit substances (eg, cocaine, methamphetamine), motor vehicle accidents, athletic injuries, permanent cognitive impairment, reduced life achievement, psychosis and/or progression to a cannabis use disorder becomes more likely.[46–50] In general, 1 out of 9 cannabis users progress to a use disorder; however, this increases to 1 in 6 if use starts in adolescence and to 1 in 4 for those that use daily.[46] Many of these adverse effects are due to the potency (THC level) of herbal US cannabis products which has risen over the past decade from approximately 10% in 2009 to 14% in 2019.[51] Similar increases have occurred in the United Kingdom, Central Europe, and New Zealand.[52] Cannabis levels are

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highest in cannabis concentrates (eg, ear wax, hash oil) with THC levels of 40% to 80%. Additional risks for concentrates include intoxication effects that are 3 to 4 times that of plant-based marijuana and the unknown risks of inhaling residual solvents.[53]

Early Detection

Annual preseason psychoeducational presentations on the patterns, prevalence, potency, and adverse effects of cannabis use and misuse to groups of athletes or teams with the distribution of a companion fact sheet or negative consequences questionnaire may prompt individuals to ask more questions about their use and review the pros and cons of continuing. Athletic trainers, team physicians, and sports mental health and performance providers should be educated regarding cannabis' early behavioral effects and typical signs and symptoms of intoxication and withdrawal (see **Table 1**).[54,55]

Random or surveillance urine testing is another way of identifying cannabis users or misusers. Random testing refers to testing small groups of athletes on a team each week with the intention of testing all team members over the season or the year. Surveillance testing refers to testing all members (or a representative sample – 50%) with or without the results being connected to a specific athlete's name to determine the point prevalence of use. This might be done for sports that are known to have the highest rates of use (eg, lacrosse, swimming, ice hockey, field hockey). High individual urine levels (ie, above 150 ng/mL) may indicate a daily or near daily user and warrant an evaluation.

Screening instruments such as the four question Cutting Down, Annoyance by Criticism, Guilty Feelings, Eye-opener modified to include cannabis/marijuana (CAGE-AID)[26] or the three question Cannabis Use Disorder Screening Tool-Short-Form (CUDIT-SF) which uses questions 3, 5, and 6 of the longer CUDIT-R[27] are recommended to identify misuse (see Appendices 2 and 3). A score of 2 or more indicates a positive screen for both the CAGE-AID and CUDIT-SF. A positive screen on one or both should result in a clinical assessment with a team physician or licensed mental health provider along with the administration of the 20-item Brief Marijuana Consequences Questionnaire.[30]

Brief Interventions

The literature on brief motivational interventions with personalized feedback for cannabis use and misuse in elite athletes is scarce. Like college students, young adult athletes at collegiate and professional levels are increasingly viewing cannabis use as safe and socially acceptable with a companion view that much higher percentages of their teammates are using than actually are.[24,25,56,57]

Given the documented adverse effects of both regular alcohol and cannabis use in athletes (see **Table 1**), brief interventions conducted by athletic trainers, team physicians, and/or sports mental health and substance providers are warranted.[24,50]

There is some evidence that brief motivational interventions with personalized feedback compared to assessment without intervention or psychoeducation alone can reduce positive expectancies and frequency of use, change attitudes about cannabis' safety and levels of normative use, and potentially prevent adverse consequences in non-athlete samples of adolescents and college students.[58–60] These motivational interventions are typically designed to encourage participants to change their behavior, teach behavioral coping and change skills, and connect to services using the FRAMES model (ie,; personalized *feedback*, *reinforcement* of personal responsibility, nonjudgmental *advice*, *menu* of strategies, *empathetic* listening, and activation of *self-efficacy*).[56]

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Parisi and colleagues (2019) in a study of 12,150 students including 1233 student athletes found sufficient adverse effects of alcohol and cannabis use to recommend increased education and brief intervention skill development for athletic personnel (eg, coaches, athletic trainers, strength and conditioning staff).[24] They developed a 2-h training module with content on patterns and prevalence of use, risks for alcohol-related injury, risks for alcohol and cannabis-related academic difficulties, and skill building for motivational interviewing. The skill-building section of the module was designed to teach athletic personnel to have "meaningful conversations" about student athlete use and facilitate guiding help-seeking behaviors.

Another intervention mentioned earlier by Rulison and colleagues (2022)[34] used an iteratively modified version of *myPlaybook* during three trials of approximately 2000 to 3000 each Division I, II, and III athletes with content on norms of alcohol and cannabis use, expectancies of drug effects, harm reduction and life skills. For cannabis, they found reductions in views of teammates' levels of use and increased negative expectancies when using.

Finally, an alternative approach by Lee and colleagues (2021)[59] developed a novel Marijuana Consequences Checklist (26 items, MCC) in college undergraduates that attempted to more accurately document the common consequences of cannabis use of typical users. They discovered that for monthly or more frequent users, the top 6 "not so good" consequences over the past 30 days were: (1) had the munchies (83.9%); (2) experienced dry mouth (3) had trouble concentrating or paying attention (62.5%); (4) acted foolish or goofy (61.9%); (5) had trouble remembering things (57.4%); and (6) had low motivation (46.4%). They suggest using the MCC to generate a motivational discussion and deliver personalized feedback based on the response patterns from the 336 undergraduates that completed the checklist.

Based on the available evidence, coaches, athletic trainers, team physicians, and/or on-site mental health or substance use providers can use team-based presentations to present information on the adverse effects of cannabis use alone or in combination with alcohol and administer either the Brief Marijuana Consequences Questionnaire[30] or the Marijuana Consequences Checklist.[61] Those with regular or heavy patterns of cannabis use and a higher number of negative consequences can then receive individual personalized feedback with an open motivational discussion of the potential negative effects of cannabis on athletic and academic performance. Those who continue the use/misuse after this "open discussion" should have a formal assessment by a mental health provider and potential referral to an outpatient treatment program.

TOBACCO/NICOTINE

Background

The use and misuse of oral (spit) tobacco and other tobacco products (eg, cigars, cigarettes, hookah) and/or nicotine vaping is regularly encountered by sports medicine providers especially those that work in baseball, hockey, wrestling, golf, and lacrosse.[16,62] Overall annual use rates of the various tobacco products and nicotine vaping for US male collegiate athletes are: oral tobacco (21.8%); cigars (17%); cigarettes (14%); hookah (10%); and electronic cigarettes (8%) with 5% reporting daily use of oral tobacco.[16] There is substantial variation in annual male use rates by sports with far higher rates for annual oral tobacco use in ice hockey (46%), baseball (44%), lacrosse (35%) and wrestling (33%) and high rates for cigarettes in lacrosse (38%), golf (23%) and ice hockey (23%). Overall annual use rates for women collegiate athletes are far lower than for men for both cigarettes (5.3% vs 14%) and oral tobacco (0.6% vs 21.8%). However, just as for men, women's use rates vary significantly by

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sport with higher use rates of cigarettes in lacrosse (10%), ice hockey (9%) tennis (9%), and golf (8%) and higher use rates for oral tobacco in ice hockey (13%) and softball (2%).[16]

One study of past 30-day use rates for 8617 collegiate athletes from the National College Health assessment II from 2011 to 2014 showed that oral tobacco use was 43% more likely in collegiate athletes for occasional use (1–19 days out of 30) and 152% more likely for frequent use (20 or more days out of 30) compared to non-athletes.[62] The same study showed that collegiate athletes were less likely to be occasional and frequent users of cigarettes and hookah (26% less for occasional hookah; 29% less for occasional cigarette use, and 52% less for frequent cigarette use) than their non-athlete counterparts. Overall male and female athlete use rates of nicotine vaping (electronic cigarettes) are daily (0.8%), weekly (0.8%), monthly (1.2%), and past year (5.4%).[16] These nicotine vaping rates however, are substantially lower than those seen in college students with 30-day use rates increasing from 6% in 2017 to 22% in 2019 and from 8% to 18% among 19-to-22-year-olds not in college.[63] The most common reason for tobacco and or nicotine vaping use in collegiate athletes are to fit in socially, for stress-relief, and to boost energy.[15] The most common adverse effects and signs and symptoms of intoxication and withdrawal are detailed in **Table 1**.[64]

There is far less information on tobacco or nicotine use among professional athletes. One study by Marclay and colleagues (2011) found overall rates in 2185 athletes from 43 different international professional sports of 15.3% using urine testing for nicotine and its metabolites.[65] When viewed by specific sports however, rates ranged from 26% to 56% depending on sport. Rates for the teams studied were as follows: American football (56%); ice hockey (32%); wrestling (32%); bobsleigh (31%); gymnastics (29%); rugby (28%); and skiing (26%).

Early Detection

Awareness of the high prevalence of tobacco use and nicotine vaping in certain sports (eg, ice hockey, baseball, American football, wrestling, golf) is one key to early detection. Questions about tobacco use and nicotine vaping should be included in the pre-season physical examination along with alcohol and cannabis. If use is endorsed, then additional screening with the 6-question Fagerstrom Dependence Scale (FDS - Appendix 4) should be considered.[66] A score of 5 or more indicates a significant dependence, while a score of 4 or less shows a low to moderate dependence. An additional approach is to have the team dentist do an oral exam at the time of the preseason physical examination looking for signs of gum irritation or oral lesions (ie, leukoplakia). If the exam is positive, then more questions about the current and past pattern of use and desire to quit should be asked.

Brief Interventions

There are no recent studies of brief interventions for elite athletes who are using or misusing tobacco products or vaping nicotine. Older studies from 1999 to 2003 in high school, collegiate, and professional baseball players (where oral tobacco use rates in minor league and major league players ranged from 25% to 35%) have demonstrated the effectiveness of brief interventions in increasing the perception of risk and reducing levels of use.[67–70] Most of these studies included a dentist or dental assistant led, peer assisted brief intervention that occurred at the time of the preseason physical exam. The intervention provided personalized feedback regarding current, or future, oral health problems, level of nicotine dependence using the FDS, and provided strategies for quitting. In some of these dental-based interventions,

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the baseball team's athletic trainers continued the follow-up by encouraging continued abstinence and/or treatment seeking with a mental health professional skilled in tobacco or nicotine vaping cessation. In one randomized controlled trial of brief intervention of 307 high school baseball players, quit rates at 2 years were 23% compared to 13% in the control group.[70] In a similar brief intervention with professional baseball players from one organization conducted over ten successive years, oral leukoplakia declined, and the prevalence of oral tobacco use fell from 22.6% in 1991 to 9.4% in 2000.[71]

Based on the available evidence there seems to be merit in a dentist/dental technician-led oral preseason evaluation along with tobacco use screenings and assessment of dependence using the FDS followed by a brief motivational intervention with a provider experienced in tobacco/nicotine cessation. Another option is group-based brief motivational intervention with a tips-for-quitting guide along with one- and 3-month follow-ups. With advanced training in this domain, athletic trainers, team physicians, and team-based mental health providers could administer this intervention and conduct the follow-up.

REFERRAL TO TREATMENT

When an athlete is not responding to the brief intervention strategies described above, or is diagnosed as having a substance use disorder, they may benefit from referral to an addiction specialist for a more comprehensive evaluation to determine the next steps. Alternatively, if the athlete is already demonstrating medical signs and symptoms of withdrawal that indicate pending instability (ie, alcohol withdrawal-**Table 1**), they may require immediate medical detoxification in an outpatient or inpatient setting.

Treatment for a substance use disorder exists on a spectrum with the American Society of Addiction Medicine (ASAM) identifying 5 benchmark levels of care as a method of standardizing addiction treatment[80]. Level 0/0.5 includes prevention and early intervention and includes brief intervention strategies discussed above. Outpatient services are the focus of Level 1, while Level 2 is comprised of intensive outpatient and partial hospitalization services. Level 3 involves residential/inpatient services including clinically managed low-intensity and high-intensity residential services, and medically monitored intensive inpatient services. Level 4 is the highest level of care and is comprised of medically managed intensive inpatient services.[72]

The topical areas explored in the criteria for determining the appropriate level of care include (but are not limited to) medical complexity, behavioral health complexity, readiness to change, environmental instability, and previous experience in treatment settings. The process of determining the level of care and criterion match is best executed by a trained clinician and is typically completed upon initial intake and evaluation to a new treatment facility. Based upon this clinical assessment, the client enters treatment at a certain level best matched to address their level of instability and need, with the goal to progress to less clinically rigorous engagement as their needs evolve. Not every program is able to satisfy the definition of the full spectrum and may offer alternative referral if the individual demonstrates needs greater than what is available.

DISCUSSION

Sports medicine physicians and athletic trainers are in prime positions to help with the early identification, assessment, and brief intervention for substance misuse and abuse among athletes. Knowing utilization and misuse patterns of the sport you are working with can help enhance awareness of symptoms guide best questions to ask. Across all sports and competition levels, the most encountered substances of

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misuse are alcohol (binge drinking), cannabis (marijuana), and nicotine (oral tobacco/nicotine vaping). While more detailed data exists for the collegiate population, there is limited data among professional athletes, especially current professional athletes in the US, making it more challenging to know prevalence rates for this population. Continued exploration of use and misuse patterns along with adverse consequences of use will help develop more effective prevention, assessment, and intervention processes for this population.

Development of an organizational plan for training, identification, intervention, and treatment prior to the start of the season will help sports medicine staff know when and how to assess for misuse of alcohol, cannabis, and nicotine. Pre-participation physicals is an ideal time for sports medicine staff to begin asking questions and assessing behaviors via self-report screeners, open-ended questioning, medical exams, and urine screens. Following transitions and acute events (ie, injury, trade, transfer, and so forth) is another time to screen for changes in behaviors and substance use patterns. While discrete transitions provide ideal times for screening, sports medicine staff have frequent interactions and access to players that allows for ongoing informal screening and behavior monitoring throughout the year.

While geared toward sports medicine physicians and athletic trainers, many other athlete-facing staff may be in roles to help facilitate prevention, identification, and early intervention depending on the organization and individual athlete relationships. Coaches, player engagement/development staff, dieticians, strength and conditioning, and equipment staff are a few of the groups that may be in positions to identify signs and symptoms of misuse/abuse. Training with all athlete-facing staff can help with identification and referral to the sports medicine staff for further assessment and evaluation when warranted.

Training for sports medicine staff on early identification, motivational interviewing, and brief interventions for alcohol, cannabis, and nicotine use is recommended. Partnering with licensed mental health providers can aid in this training and provide an easy transition from brief intervention to higher levels of ongoing care. The ASAM levels of care outline the continuum of treatment based on patterns of use and symptoms. It is important for sports medicine staff to develop relationships with addiction specialists in the community along with local area treatment facilities providing the full spectrum of care. These relationships will help facilitate access to, and transition of care when higher levels of treatment is warranted.

CLINICS CARE POINTS

- Screening for alcohol, cannabis, and tobacco/nicotine misuse prior to each season can lead to early detection and brief intervention.
- The simultaneous use of alcohol with either cannabis or tobacco/nicotine is associated with increased adverse effects.
- Brief Motivational Intervention training for substance misuse should become a part of the ongoing education of athletic trainers and team physicians.

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APPENDIX

AUDIT- C

The following questions are about alcohol use. Please respond to each question by circling the number from "0" to "4" that represents your alcohol use.

1. How often do you have a drink containing alcohol?
   Never (0) monthly or less (1) 2 to 4 times a month (2) 2 to 3 times a week (3) 4 or more times a week (4)
2. How many standard drinks containing alcohol do you have on a typical day when you drink?
   1 to 2 (0) 3 to 4 (1) 5 to 6 (2) 7 to 9 (3) 10 or more (4)
3. How often do you have six or more drinks on one occasion?
   Never (0) less than monthly (1) monthly (2) weekly (3) daily or almost daily (4)

Calculate the score by summing up the answers on the 3 items. A score of $\geq$4 for men & $\geq$3 for women is a positive screen.

CAGE-AID (alcohol, cannabis/marijuana, tobacco/nicotine, other drug(s))

The following questions are about alcohol, cannabis/marijuana tobacco/nicotine use in the last 3 months. Please respond to each question by circling "yes" or "no."

1. In the last 3 months, have you felt you should cut down or stop using alcohol, cannabis, tobacco/nicotine or other drugs?
   Alcohol Yes (1) No (0); *Cannabis* Yes (1) No (0) *Tobacco/nicotine* Yes (1) No (0) *Other drugs* Yes (1) No(0)
2. In the last 3 months, has anyone annoyed you or gotten on your nerves by telling you to cut down or stop using alcohol, cannabis, tobacco/nicotine, other drugs ?
   Alcohol Yes (1) No (0); *Cannabis* Yes (1) No (0) *Tobacco/nicotine* Yes (1) No (0) *Other drugs* Yes (1) No(0)
3. In the last 3 months, have you felt guilty or bad about how much you use alcohol, cannabis, tobacco/nicotine, other drugs?
   Alcohol Yes (1) No (0); *Cannabis* Yes (1) No (0) *Tobacco/nicotine* Yes (1) No (0) *Other drugs* Yes (1) No(0)
4. In the last 3 months, have you been waking up wanting to use alcohol, cannabis, tobacco/nicotine, other drugs?
   Alcohol Yes (1) No (0); *Cannabis* Yes (1) No (0) *Tobacco/nicotine* Yes (1) No (0) *Other drugs* Yes (1) No(0)

Calculate the total score for each substance by summing up the answers on the four items. A score of >2 is a positive screen for men and women.

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CUDIT-SF

Do you currently use cannabis? YES or NO; If YES

1. How often during the past 6 months did you find that you were not able to stop using cannabis once you had started?
   Never (0) Less than monthly (1) monthly (2) Weekly (3) Daily or almost daily (4)
2. How often in the past 6 months have you devoted a great deal of time getting, using, or recovering from cannabis?
   Never (0) Less than monthly (1) monthly (2) Weekly (3) Daily or almost daily (4)
3. How often in the past 6 months have you had a problem with you memory or concentration after using cannabis?
   Never (0) Less than monthly (1) monthly (2) Weekly (3) Daily or almost daily (4)

Calculate the total score by summing up the answers on the 3 items. A score of >2 is a positive screen for women and men.

Fagerstrom Test for Nicotine Dependence

Do you currently use tobacco products? No Yes

If "yes," read each question listed. For each question, circle the answer choice which best describes your response.

1. How soon after you wake up do you smoke your first cigarette?
   Within 5 minutes (3) 6 to 30 minutes (2) 31 to 60 minutes (1) After 60 minutes (0)
2. Do you find it difficult to refrain from smoking in places where it is forbidden (for example, in church, at the library, in the cinema)? No (0) Yes (1)
3. Which cigarette would you hate most to give up?
   The first one in the morning (1) Any other (0)
4. How many cigarettes per day do you smoke?
   10 or less (0) 11 to 20 (1) 21 to 30 (2) 31 or more (3)
5. Do you smoke more frequently during the first hours after waking than during the rest of the day? No (0) Yes (1)
6. Do you smoke when you are so ill that you are in bed most of the day? No (0) Yes (1)

Scoring Instructions: Add up the responses to all items. A score of 5 or more indicates a significant dependence, while a score of 4 or less shows a low to moderate dependence.

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